Huawei Access Node Routing Protocol Flap Resolution Procedure for BTM-ACC-34

Node: BTM-ACC-34 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_6091

Description

The Routing Protocol Flap failure domain, denoted by the critical alarm code ALARM_6091 on the Huawei access node BTM-ACC-34, situated at the BTM Layout Aggregation site, is a complex issue stemming from the intricacies of routing protocol dynamics. Specifically, this failure can be attributed to the instability in Border Gateway Protocol (BGP) route churn, Open Shortest Path First (OSPF) adjacency loss, or Intermediate System to Intermediate System (ISIS) Link-State Protocol (LSP) flooding. These routing protocols are fundamental to the establishment and maintenance of network topology and reachability information across the network. The BGP route churn could be due to instability in the network, causing frequent updates and withdrawals of routing information, potentially leading to routing loops or black holes. OSPF adjacency loss indicates a problem in the formation or maintenance of neighbor relationships between routers, critical for the exchange of link-state information. ISIS LSP flooding, on the other hand, could be a result of excessive link-state updates, possibly triggered by network topology changes or configuration issues. Given the critical nature of this failure and its potential to disrupt network services, a meticulous and step-by-step approach is required to diagnose and resolve the issue, ensuring minimal impact on the network and its users.

Pre-requisite Checks

- Verify the physical integrity of the node and its connections at the BTM Layout Aggregation site.

- Check the status of redundant links to ensure they are operational and capable of taking over in case of a failure.

- Ensure that all relevant network management systems and monitoring tools are accessible and providing real-time data.

- Confirm the availability of remote hands or on-site support for immediate intervention if required.

Potential Root Causes

Cause	Probability
BGP Route Churn Due to Network Instability	High
OSPF Adjacency Loss Due to Configuration Issues	Medium
ISIS LSP Flooding Caused by Excessive Topology Changes	Low

Diagnostic Commands & Expected Outputs

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Command: display ip routing-table, Expected Output: Routing table
information showing prefixes, next hops, and routing protocols.,
Description: This command isolates issues in the control plane by
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displaying the current routing table, allowing for the verification of routing information and potential inconsistencies that could indicate a routing protocol flap.

Command: display bgp peer, Expected Output: BGP peer information including state, address, and description., Description: This command provides insight into the BGP peer relationships, helping to identify any issues with BGP session establishment or maintenance that could contribute to route churn.

Command: display ospf peer, Expected Output: OSPF neighbor information detailing state, interface, and IP address., Description: This command is used to diagnose OSPF adjacency issues by examining the state of OSPF neighbors, which is crucial for link-state information exchange.
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Resolution Steps

Step 1: Step: Adjust BGP Route Reflectors, Command: bgp router-reflector cluster-id , Description: Configure BGP route reflectors to optimize route reflection and reduce the likelihood of route churn. This involves specifying the autonomous system number and configuring the cluster ID for route reflection.

Step 2: Step: Modify OSPF Interface Costs, Command: ospf cost interface , Description: Adjust the OSPF interface costs to influence the selection of paths and mitigate adjacency loss. This step requires careful analysis of the network topology to determine the optimal costs.

Step 3: Step: Tune ISIS LSP Flooding Parameters, Command: isis timer lsp-generation , Description: Configure the ISIS LSP flooding parameters to control the frequency of LSP updates. This helps in preventing excessive flooding that could destabilize the network.

Post-Check Verification

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Command: display ip routing-table verbose, Expected Output: Detailed routing table information including protocol, prefix, next hop, and interface., Description: This command is used to verify that the routing table has stabilized and reflects the expected changes made during the resolution steps, indicating successful resolution of the routing protocol flap.

Command: display bgp routing-table, Expected Output: BGP routing table showing learned routes and their attributes., Description: Verify that BGP routes are being correctly learned and advertised, confirming the stability of BGP sessions and route churn mitigation.
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Roll-back Procedures

- Step: Revert BGP Configuration, Command: undo bgp router-reflector, Description: If the adjustments to BGP route reflectors do not yield the desired outcome or introduce unforeseen issues, this step involves reverting the BGP configuration to its previous state to maintain network stability.

- Step: Restore OSPF Interface Costs, Command: undo ospf cost interface , Description: In case the modifications to OSPF interface costs do not resolve the adjacency loss or introduce suboptimal routing, this step restores the original OSPF costs to prevent further disruptions.